

unite: Unified liNe Integration Turbo Engine

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Summary

Astronomical spectroscopy, whereby light from celestial sources is dispersed into its constituent wavelengths (colors), is a cornerstone of modern astrophysical research. In particular the characterization of spectral lines, which arise from atomic and molecular transitions in astrophysical gas, allows astronomers to measure fundamental physical properties such as redshift, chemical composition, temperature, density, and kinematics of the emitting or absorbing material. Therefore, the flexibility, speed, and accuracy of spectral line-fitting tools directly impact the scientific return of spectroscopic observations.

unite (Unified liNe Integration Turbo Engine) is a Python package for fast and accurate Bayesian inference of spectral line features from one or more spectra simultaneously. It is built primarily on JAX ([Bradbury et al., 2018](#)) (for speed and automatic differentiation), NumPyro ([Bingham et al., 2019](#); [Phan et al., 2019](#)) (for probabilistic programming), and Astropy ([Astropy Collaboration et al., 2013, 2018, 2022](#)) (for units and FITS handling), and is designed to be flexible and extensible to a wide range of spectroscopic applications.

Statement of need

The Near-Infrared Spectrograph (NIRSpec; Jakobsen et al. ([2022](#))) on the James Webb Space Telescope (JWST; Gardner et al. ([2023](#))) is the frontier instrument for near-infrared spectroscopy and has already provided revolutionary insights across a wide range of astrophysical topics, from the characterization of exoplanet atmospheres to the discovery of the most distant galaxies known. However, a defining challenge of NIRSpec spectroscopy is that the detector critically undersamples the instrumental line spread function (LSF) across all gratings and observing modes. Therefore, evaluating the model at pixel centers rather than integrating over the pixel domains introduces systematic errors in recovered fluxes and line widths, which can bias scientific inferences. When existing spectral line fitting tools can account for this, they typically do so via computationally and memory expensive super-sampling and convolution. In addition, typical JWST/NIRSpec programs observe targets with multiple gratings to leverage their complementary strengths (i.e. high-resolution to measure kinematics, low-resolution to constrain fluxes and continuum shapes), so a statistically optimal analysis must fit all gratings simultaneously with shared astrophysical parameters and grating-specific calibration offsets. unite is designed to address both of these challenges while additionally providing a flexible, extensible framework for spectroscopic analysis that can be applied to any instrument with user-supplied resolving power and pixel-scale functions. By relying on existing optimized libraries for probabilistic programming and automatic differentiation, unite delivers scientifically rigorous Bayesian inference without compromising on speed, enabling the analysis of large spectroscopic datasets with accurate uncertainty quantification. The target audience is observational astronomers, especially those working with JWST/NIRSpec data, that want to do inference over large spectroscopic samples, deal with undersampled data, and/or fit multiple spectra simultaneously with shared parameters.

43 Software design

44 When fitting spectroscopic data sets, fitting routines typically assume that the model evaluated
45 at the pixel center is a good representation of the average of the model over the pixel domain,
46 which is what the instrument actually measures. This approximation is well justified when the
47 spectrum is critically sampled or over-sampled, i.e. when the signal changes slowly over the
48 pixel domain, but breaks down when the spectrum is undersampled and the signal changes
49 rapidly over the pixel domain, as is the case for NIRSpec. This can be addressed integrating the
50 model over the pixel domain, providing the exact solution for the observed signal regardless of
51 the degree of undersampling. We rely on the assumption that the LSF is well approximated by
52 a Gaussian kernel, which is a good approximation for NIRSpec and many other spectrographs,
53 especially in the undersampled regime.

54 `unite` computes the integrals of continua and line models analytically where possible. However,
55 analytic pixel integration is not possible for all model setups, in particular in the presence of
56 optical-depth parametrized absorption lines where the nonlinear transmission $e^{-\tau\phi}$ couples the
57 line depth and profile shape in a way that prevents closed-form solutions for the pixel integrals.
58 In these cases, `unite` provides a numerical convolution mode which supersamples the intrinsic
59 model on a fine wavelength grid and convolves with the wavelength-dependent LSF kernel to
60 produce a pixel-convolved model. This model accurately captures the nonlinear coupling of line
61 depth and profile shape, but is more computationally expensive than the analytic integration
62 mode; users can choose the appropriate mode for their application.

63 Despite implementing integration, quadrature, and convolution modes, `unite` is fast and
64 efficient thanks to its JAX backend, which provides just-in-time (JIT) compilation and native
65 GPU support. At its core, `unite` is a domain-specific language for building probabilistic
66 models of spectroscopic data. Users build a declarative configuration of line and continuum
67 components and assign priors to physical parameters via token instances which can be shared
68 across multiple model components with arithmetic combinations. The prior system is expressive:
69 priors on any parameter can depend on other parameters through arithmetic expressions and
70 topologically sorted dependency chains, enabling physical constraints such as requiring a broad
71 component's velocity width to exceed that of a narrow component by a certain amount, or
72 fixing flux ratios between doublet lines.

73 In addition, users specify the instrumental configuration carrying empirical calibrations of the
74 wavelength-dependent resolving power, pixel scale, and flux normalization for each disperser,
75 which can be shared across instruments. One aspect that sets `unite` apart from other spectral
76 fitting tools is that it treats instrumental calibration parameters as first-class citizens in the
77 inference process; pixel offsets, resolution scales, and flux normalizations can be directly
78 incorporated into the model with priors and sampled jointly with astrophysical parameters,
79 allowing for instrumental uncertainties to directly propagate to the inferred properties. For
80 example, when fitting NIRSpec data, users can fold in the empirically observed offsets in
81 wavelength solutions and flux scales derived in de Graaff, Brammer, et al. (2025) to obtain
82 realistic uncertainties on line fluxes and kinematics that account for systematic uncertainties in
83 the instrument calibration.

84 All configurations are serializable to human-readable YAML for reproducibility and sharing.
85 `unite` assembles a NumPyro probabilistic model for inference with any compatible sampler,
86 including SVI for quick exploratory fits, NUTS for full posterior sampling, and nested sampling
87 for model comparison and evidence calculation. Finally, `unite` provides convenience functions
88 for extracting results as parameter tables and per-spectrum model predictions into domain-
89 appropriate FITS HDU lists.

90 The package is publicly available on GitHub and PyPI under the GPL-3.0-or-later license,
91 with a DOI minted via Zenodo (Hviding, 2026) and accompanied by Sphinx documentation
92 including narrative guides, an API reference, and executable tutorials. CI/CD workflows ensure
93 that the code is tested and documented with each update, and the project is open to feedback

94 and contributions from the community.

95 State of the field

96 Spectral line analysis is among the most common operations in observational astronomy, and
97 the landscape of fitting software is correspondingly rich.

98 The dominant paradigm for emission-line fitting is non-linear least squares, with most Python
99 codes built on LMFIT (Newville et al., 2014) or `scipy.optimize` (Virtanen et al., 2020) directly.

100 pPXF (Cappellari, 2017, 2023; Cappellari & Emsellem, 2004) is the standard tool for stellar
101 kinematics and stellar-population recovery, fitting observed galaxy spectra as non-negative linear
102 combinations of SSP templates and emission lines convolved with a parametric line-of-sight
103 velocity distribution with optional supersampling.

104 PySpecKit (Ginsburg et al., 2022) is a general-purpose spectral fitting toolkit supporting a
105 range of profile shapes and optional MCMC uncertainty estimation.

106 LiMe (Fernández et al., 2024) is a modern library designed for large and complex datasets
107 including JWST spectra, with batch fitting across many spectra and flexible profile shapes.

108 Q3DFIT (Rupke, 2014; Rupke et al., 2021, 2023) targets IFU spectroscopy of quasar host
109 galaxies, fitting multi-component Gaussian profiles via least squares with the stellar continuum
110 pre-subtracted using pPXF.

111 BADASS (Sexton et al., 2021, 2024) is a comprehensive Bayesian emission-line code to
112 simultaneously infer AGN power-law continuum, FeII pseudo-continuum, stellar LOSVD (via
113 pPXF), and multi-component Gaussian emission lines in a single posterior.

114 These codes occupy various niches but none provide pixel integration or first-class joint multi-
115 grating fitting. In addition, the fitting paradigm for many of these codes is least-squares
116 optimization, which does not provide the posterior distribution over parameters. For those
117 that do provide Bayesian inference, they typically rely on `emcee` (Foreman-Mackey et al., 2013)
118 or `pymc3` (Abril-Pla et al., 2023) which can struggle in high-dimensional parameter spaces and
119 do not natively support (GPU) acceleration. However many incorporate templates into their
120 fitting frameworks, which `unite` does not currently support.

121 Research impact statement

122 `unite` has already been used in several published JWST/NIRSpec analyses, demonstrating its
123 utility for emission line characterization.

- 124 ■ **Accurate Line Fluxes and Kinematics:** With the accurate accounting for undersampling,
125 `unite` has been used to robustly characterize fluxes and kinematics for both emission
126 lines and absorption features in high-redshift galaxies, providing insights into the physical
127 processes driving star formation and black hole growth. (de Graaff, Hviding, et al., 2025;
128 Naidu et al., 2025; Sun et al., 2026; Wang et al., 2025, 2026)
- 129 ■ **Multi-Component Line Fitting:** `unite` has been used to identify broad Balmer emission
130 (broad $H\alpha$, $H\beta$) in high-redshift ($z \gtrsim 4$) galaxies observed with NIRSpec, providing
131 evidence for active supermassive black hole growth in the early universe. Multi-grating
132 joint fitting is critical as grating spectroscopy often lacks in signal-to-noise but by coupling
133 physical constraints on flux from low-resolution gratings, `unite` enables robust detections
134 of these components. (de Graaff, Hviding, et al., 2025; Hviding et al., 2025, 2026)
- 135 ■ **Redshift Precision:** `unite` was used in the analysis of MoM-z14, which at the time
136 of publication was (and remains) the most distant spectroscopically confirmed galaxy
137 known. Coupling line parameters and properly accounting for undersampling allowed for

a factor of 5 improvement in redshift precision than from continuum estimates alone even with faint, marginally detected, emission lines. (Naidu et al., 2026)

AI usage disclosure

Claude Code (Anthropic) was used as an AI coding assistant during the development of unite, its documentation, and the drafting of this paper. It was used to generate new code, refactor existing code, and write certain sections of the documentation. All scientific decisions, algorithmic design choices, and final content were made and verified by the authors.

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References

- Abril-Pla, O., Andreani, V., Carroll, C., Dong, L., Fonnesbeck, C. J., Kochurov, M., Kumar, R., Lao, J., Luhmann, C. C., Martin, O. A., Osthege, M., Vieira, R., Wiecki, T., & Zinkov, R. (2023). PyMC: A modern and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*, 9(e1516). <https://doi.org/10.7717/peerj-cs.1516>
- Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., Earl, N., Starkman, N., Bradley, L., Shupe, D. L., Patil, A. A., Corrales, L., Brasseur, C. E., Nöthe, M., Donath, A., Tollerud, E., Morris, B. M., Ginsburg, A., Vaher, E., Weaver, B. A., Tocknell, J., Jamieson, W., ... Astropy Project Contributors. (2022). The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. 935(2), 167. <https://doi.org/10.3847/1538-4357/ac7c74>
- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., Günther, H. M., Lim, P. L., Crawford, S. M., Conseil, S., Shupe, D. L., Craig, M. W., Dencheva, N., Ginsburg, A., VanderPlas, J. T., Bradley, L. D., Pérez-Suárez, D., de Val-Borro, M., Paper Contributors, (., Aldcroft, T. L., Cruz, K. L., Robitaille, T. P., ... Contributors, (. (2018). The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package. 156, 123. <https://doi.org/10.3847/1538-3881/aabc4f>
- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., Greenfield, P., Droettboom, M., Bray, E., Aldcroft, T., Davis, M., Ginsburg, A., Price-Whelan, A. M., Kerzendorf, W. E., Conley, A., Crighton, N., Barbary, K., Muna, D., Ferguson, H., Grollier, F., Parikh, M. M., Nair, P. H., ... Streicher, O. (2013). Astropy: A community Python package for astronomy. 558, A33. <https://doi.org/10.1051/0004-6361/201322068>
- Bingham, E., Chen, J. P., Jankowiak, M., Obermeyer, F., Pradhan, N., Karaletsos, T., Singh, R., Szerlip, P. A., Horsfall, P., & Goodman, N. D. (2019). Pyro: Deep universal probabilistic programming. *J. Mach. Learn. Res.*, 20, 28:1–28:6. <http://jmlr.org/papers/v20/18-403.html>
- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Katariya, Y., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018). JAX: Composable transformations of Python+NumPy programs (Version 0.3.13). <http://github.com/google/jax>
- Cappellari, M. (2017). Improving the full spectrum fitting method: accurate convolution with Gauss-Hermite functions. *MNRAS*, 466(1), 798–811. <https://doi.org/10.1093/mnras/stw3020>

- 181 Cappellari, M. (2023). Full spectrum fitting with photometry in PPXF: stellar population
182 versus dynamical masses, non-parametric star formation history and metallicity for 3200
183 LEGA-C galaxies at redshift $z \approx 0.8$. *MNRAS*, 526(3), 3273–3300. <https://doi.org/10.1093/mnras/stad2597>
184
- 185 Cappellari, M., & Emsellem, E. (2004). Parametric Recovery of Line-of-Sight Velocity
186 Distributions from Absorption-Line Spectra of Galaxies via Penalized Likelihood. *PASP*,
187 116(816), 138–147. <https://doi.org/10.1086/381875>
- 188 de Graaff, A., Brammer, G., Weibel, A., Lewis, Z., Maseda, M. V., Oesch, P. A., Bezanson, R.,
189 Boogaard, L. A., Cleri, N. J., Cooper, O. R., Gottumukkala, R., Greene, J. E., Hirschmann,
190 M., Hviding, R. E., Katz, H., Labbé, I., Leja, J., Matthee, J., McConachie, I., ... Williams,
191 C. C. (2025). RUBIES: A complete census of the bright and red distant Universe with
192 JWST/NIRSpec. *A&A*, 697, A189. <https://doi.org/10.1051/0004-6361/202452186>
- 193 de Graaff, A., Hviding, R. E., Naidu, R. P., Greene, J. E., Miller, T. B., Leja, J., Matthee,
194 J., Brammer, G., Katz, H., Bezanson, R., Boogaard, L. A., Bose, S., Chisholm, J.,
195 Cleri, N. J., Dayal, P., Feldmann, R., Fudamoto, Y., Fujimoto, S., Furtak, L. J., ...
196 van der Wel, A. (2025). Little Red Dots host Black Hole Stars: A unified family of
197 gas-reddened AGN revealed by JWST/NIRSpec spectroscopy. *arXiv*, arXiv:2511.21820.
198 <https://doi.org/10.48550/arXiv.2511.21820>
- 199 Fernández, V., Amorín, R., Firpo, V., & Morisset, C. (2024). LIME: A Line MEasuring library
200 for large and complex spectroscopic data sets. I. Implementation of a virtual observatory
201 for JWST spectra. *A&A*, 688, A69. <https://doi.org/10.1051/0004-6361/202449224>
- 202 Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. (2013). emcee: The MCMC
203 Hammer. *PASP*, 125(925), 306. <https://doi.org/10.1086/670067>
- 204 Gardner, J. P., Mather, J. C., Abbott, R., Abell, J. S., Abernathy, M., Abney, F. E., Abraham,
205 J. G., Abraham, R., Abul-Huda, Y. M., Acton, S., Adams, C. K., Adams, E., Adler, D.
206 S., Adriaensen, M., Aguilar, J. A., Ahmed, M., Ahmed, N. S., Ahmed, T., Albat, R., ...
207 Dellagatta, G. (2023). The James Webb Space Telescope Mission. *PASP*, 135(1048),
208 068001. <https://doi.org/10.1088/1538-3873/acd1b5>
- 209 Ginsburg, A., Sokolov, V., de Val-Borro, M., Rosolowsky, E., Pineda, J. E., Sipőcz, B. M., &
210 Henshaw, J. D. (2022). Pyspeckit: A Spectroscopic Analysis and Plotting Package. *AJ*,
211 163(6), 291. <https://doi.org/10.3847/1538-3881/ac695a>
- 212 Hviding, R. E. (2026). *Unite: Unified liNe integration turbo engine* (Version v2.9.2). Zenodo.
213 <https://doi.org/10.5281/zenodo.20158737>
- 214 Hviding, R. E., de Graaff, A., Liu, H., Goulding, A. D., Ma, Y., Greene, J. E., Boogaard, L.
215 A., Bunker, A. J., Cleri, N. J., Franx, M., Hirschmann, M., Leja, J., Matthee, J., Naidu,
216 R. P., Setton, D. J., Übler, H., Venturi, G., & Wang, B. (2026). The X-Ray Dot: Exotic
217 Dust or a Late-stage Little Red Dot? *ApJL*, 1000(1), L18. <https://doi.org/10.3847/2041-8213/ae4c88>
218
- 219 Hviding, R. E., de Graaff, A., Miller, T. B., Setton, D. J., Greene, J. E., Labbé, I., Brammer,
220 G., Bezanson, R., Boogaard, L. A., Cleri, N. J., Leja, J., Maseda, M. V., McConachie, I.,
221 Matthee, J., Naidu, R. P., Oesch, P. A., Wang, B., Whitaker, K. E., & Williams, C. C.
222 (2025). RUBIES: A spectroscopic census of little red dots: All point sources with v-shaped
223 continua have broad lines. *A&A*, 702, A57. <https://doi.org/10.1051/0004-6361/202555816>
- 224 Jakobsen, P., Ferruit, P., Alves de Oliveira, C., Arribas, S., Bagnasco, G., Barho, R., Beck, T. L.,
225 Birkmann, S., Böker, T., Bunker, A. J., Charlot, S., de Jong, P., de Marchi, G., Ehrenwinkler,
226 R., Falcolini, M., Fels, R., Franx, M., Franz, D., Funke, M., ... Zincke, C. (2022). The
227 Near-Infrared Spectrograph (NIRSpec) on the James Webb Space Telescope. I. Overview
228 of the instrument and its capabilities. *A&A*, 661, A80. <https://doi.org/10.1051/0004-6361/202142663>
229

- 230 Naidu, R. P., Matthee, J., Katz, H., de Graaff, A., Oesch, P., Smith, A., Greene, J. E.,
231 Brammer, G., Weibel, A., Hviding, R., Chisholm, J., Labbé, I., Simcoe, R. A., Witten,
232 C., Atek, H., Baggen, J. F. W., Belli, S., Bezanson, R., Boogaard, L. A., ... Watson, D.
233 (2025). A “Black Hole Star” Reveals the Remarkable Gas-Enshrouded Hearts of the Little
234 Red Dots. *arXiv*, arXiv:2503.16596. <https://doi.org/10.48550/arXiv.2503.16596>
- 235 Naidu, R. P., Oesch, P. A., Brammer, G., Weibel, A., Li, Y., Matthee, J., Chisholm, J., Pollock,
236 C. L., Heintz, K. E., Johnson, B. D., Shen, X., Hviding, R. E., Leja, J., Tacchella, S.,
237 Ganguly, A., Witten, C., Atek, H., Belli, S., Bose, S., ... Whitaker, K. E. (2026). A Cosmic
238 Miracle: A Remarkably Luminous Galaxy at $z_{\text{spec}} = 14.44$ Confirmed with JWST. *OJAp*,
239 9, 56033. <https://doi.org/10.33232/001c.156033>
- 240 Newville, M., Stensitzki, T., Allen, D. B., & Ingargiola, A. (2014). *LMFIT: Non-linear*
241 *least-square minimization and curve-fitting for python* (Version 0.8.0). Zenodo. <https://doi.org/10.5281/zenodo.11813>
- 243 Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated
244 probabilistic programming in NumPyro. *arXiv Preprint arXiv:1912.11554*.
- 245 Rupke, D. S. N. (2014). *IFSFIT: Spectral Fitting for Integral Field Spectrographs* (p.
246 ascl:1409.005).
- 247 Rupke, D. S. N., Schweitzer, M., Viola, V., Lutz, D., Sturm, E., Spoon, H., Veilleux, S., &
248 Kim, D.-C. (2021). *QUESTFIT: Fitter for mid-infrared galaxy spectra* (p. ascl:2112.002).
- 249 Rupke, D. S. N., Wylezalek, D., Zakamska, N., Veilleux, S., Vayner, A., Bertemes, C., Ishikawa,
250 Y., Liu, W., Lim, H. X. G., Murphree, G., Whitesell, L., McCrory, R., & Anicetti, C. (2023).
251 *q3dfit: PSF decomposition and spectral analysis for JWST-IFU spectroscopy*. Astrophysics
252 Source Code Library, record ascl:2310.004.
- 253 Sexton, R. O., Doan, S. M., Reefe, M., & Matzko, W. (2024). *BADASS: Bayesian AGN*
254 *Decomposition Analysis for SDSS Spectra*. Astrophysics Source Code Library, record
255 ascl:2412.004.
- 256 Sexton, R. O., Matzko, W., Darden, N., Canalizo, G., & Gorjian, V. (2021). Bayesian
257 AGN Decomposition Analysis for SDSS spectra: a correlation analysis of [O III] $\lambda 5007$
258 outflow kinematics with AGN and host galaxy properties. *MNRAS*, 500(3), 2871–2895.
259 <https://doi.org/10.1093/mnras/staa3278>
- 260 Sun, W. Q., Naidu, R. P., Matthee, J., de Graaff, A., Chisholm, J., Greene, J. E., Oesch,
261 P. A., Torralba, A., Hviding, R. E., Brammer, G., Simcoe, R. A., Bose, S., Bouwens,
262 R., Dayal, P., Eilers, A.-C., Fei, Q., Furtak, L. J., Gottumukkala, R., Goulding, A., ...
263 Williams, C. C. (2026). Little Red Dot — Host Galaxy = Black Hole Star: A Gas-
264 Enshrouded Heart at the Center of Every Little Red Dot. *arXiv*, arXiv:2601.20929. <https://doi.org/10.48550/arXiv.2601.20929>
- 266 Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D.,
267 Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson,
268 J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy
269 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in
270 Python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- 271 Wang, B., Leja, J., Katz, H., Inayoshi, K., Cleri, N. J., de Graaff, A., Hviding, R. E., van
272 Dokkum, P., Greene, J. E., Labbé, I., Matthee, J., McConachie, I., Naidu, R. P., & Nelson,
273 E. J. (2025). The Missing Hard Photons of Little Red Dots: Their Incident Ionizing Spectra
274 Resemble Massive Stars. *arXiv*, arXiv:2508.18358. [https://doi.org/10.48550/arXiv.2508.](https://doi.org/10.48550/arXiv.2508.18358)
275 [18358](https://doi.org/10.48550/arXiv.2508.18358)
- 276 Wang, B., Leja, J., Labbe, I., Greene, J. E., Liu, H., de Graaff, A., Hviding, R. E., Matthee,
277 J., Quataert, E., Bezanson, R., Boogaard, L. A., Brammer, G., Burgasser, A. J., Chen,
278 Y.-X., Cleri, N. J., Cutler, S. E., Dayal, P., Furtak, L. J., Fujimoto, S., ... Zitrin, A. (2026).

279 Water absorption confirms cool atmospheres in two little red dots. *arXiv*, arXiv:2602.06024.
280 <https://doi.org/10.48550/arXiv.2602.06024>

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